

TABLE I: Detailed case-level results of MMO_b , BOHNAS, RS for neural architecture search, showing Scott-Knott ESD ranks and the mean $\pm$ standard deviation of optimized and normalized performance (smaller is better). Blue denote the best optimizer for each SE case.

case	Rank			Mean $\pm$ Std		
	MMO_b	BOHNAS	RS	MMO_b	BOHNAS	RS
c10mop10_0	1.0	2.0	1.0	0.164 ± 0.088	0.421 ± 0.339	0.292 ± 0.217
c10mop11_0	1.0	3.0	2.0	0.438 ± 0.000	0.582 ± 0.299	0.450 ± 0.141
c10mop12_0	1.0	1.0	2.0	0.629 ± 0.000	0.543 ± 0.311	0.607 ± 0.057
c10mop13_0	3.0	1.0	2.0	0.919 ± 0.096	0.760 ± 0.236	0.793 ± 0.299
c10mop1_0	1.0	3.0	2.0	0.400 ± 0.378	0.815 ± 0.157	0.540 ± 0.388
c10mop3_0	1.0	3.0	2.0	0.100 ± 0.316	0.600 ± 0.516	0.332 ± 0.366
c10mop5_0	1.0	3.0	2.0	0.000 ± 0.000	0.312 ± 0.391	0.089 ± 0.274
c10mop8_0	2.0	1.0	3.0	0.342 ± 0.159	0.285 ± 0.178	0.756 ± 0.187
citysegmop3_0	2.0	2.0	1.0	0.253 ± 0.139	0.467 ± 0.322	0.355 ± 0.177
in1kmop1_0	3.0	1.0	2.0	0.786 ± 0.192	0.160 ± 0.111	0.449 ± 0.169
in1kmop4_0	2.0	1.0	3.0	0.407 ± 0.220	0.313 ± 0.144	0.628 ± 0.184
in1kmop7_0	2.0	1.0	3.0	0.277 ± 0.106	0.151 ± 0.108	0.695 ± 0.173
Average	1.67	1.83	2.08	0.393 ± 0.141	0.451 ± 0.259	0.499 ± 0.219

TABLE II: Detailed case-level results of MMO_b , BMA, RS for next release problem. The formate is the same as Table I.

case	Rank			Mean $\pm$ Std		
	MMO_b	BMA	RS	MMO_b	BMA	RS
nrp-e1	1.0	2.0	3.0	0.002 ± 0.002	0.468 ± 0.027	0.988 ± 0.010
nrp-e2	1.0	2.0	3.0	0.001 ± 0.002	0.468 ± 0.035	0.964 ± 0.024
nrp-e3	1.0	2.0	3.0	0.000 ± 0.000	0.450 ± 0.029	0.957 ± 0.021
nrp-e4	1.0	2.0	3.0	0.000 ± 0.002	0.442 ± 0.041	0.967 ± 0.031
nrp-g1	1.0	2.0	3.0	0.000 ± 0.000	0.431 ± 0.030	0.973 ± 0.025
nrp-g2	1.0	2.0	3.0	0.000 ± 0.000	0.362 ± 0.048	0.973 ± 0.026
nrp-g3	1.0	2.0	3.0	0.000 ± 0.000	0.389 ± 0.046	0.965 ± 0.030
nrp-g4	1.0	2.0	3.0	0.000 ± 0.000	0.375 ± 0.039	0.967 ± 0.031
nrp-m1	1.0	2.0	3.0	0.001 ± 0.002	0.555 ± 0.040	0.974 ± 0.018
nrp-m2	1.0	2.0	3.0	0.002 ± 0.001	0.531 ± 0.048	0.978 ± 0.018
nrp-m3	1.0	2.0	3.0	0.001 ± 0.001	0.556 ± 0.028	0.978 ± 0.018
nrp-m4	1.0	2.0	3.0	0.001 ± 0.001	0.511 ± 0.029	0.977 ± 0.013
nrp1	1.0	2.0	3.0	0.000 ± 0.000	0.056 ± 0.029	0.938 ± 0.050
nrp2	1.0	2.0	3.0	0.029 ± 0.026	0.521 ± 0.061	0.972 ± 0.014
nrp3	1.0	2.0	3.0	0.003 ± 0.003	0.292 ± 0.029	0.979 ± 0.017
nrp4	1.0	2.0	3.0	0.000 ± 0.000	0.340 ± 0.029	0.988 ± 0.009
nrp5	1.0	2.0	3.0	0.000 ± 0.000	0.265 ± 0.023	0.987 ± 0.010
Average	1.00	2.00	3.00	0.002 ± 0.002	0.412 ± 0.036	0.972 ± 0.021

TABLE III: Detailed case-level results of MMO_b , PromiseTune, and RS for software configuration tuning. $\times$ indicates that the algorithm failed to obtain valid solutions on that case. The formate is the same as Table I.

case	Rank			Mean $\pm$ Std		
	MMO_b	PromiseTune	RS	MMO_b	PromiseTune	RS
dnn_adiaac	1.0	2.0	2.0	0.471 ± 0.379	0.522 ± 0.490	0.820 ± 0.142
dnn_adiaac_reverse	1.0	2.0	2.0	0.002 ± 0.006	0.100 ± 0.316	0.094 ± 0.034
dnn_coffee	2.0	3.0	1.0	0.314 ± 0.409	0.800 ± 0.310	0.214 ± 0.345
dnn_coffee_reverse	1.0	2.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.552 ± 0.281
dnn_dsr	1.0	2.0	2.0	0.102 ± 0.198	0.415 ± 0.398	0.113 ± 0.232
dnn_dsr_reverse	1.0	2.0	3.0	0.000 ± 0.000	0.217 ± 0.286	0.378 ± 0.271
dnn_sa	2.0	1.0	3.0	0.223 ± 0.204	0.223 ± 0.416	0.622 ± 0.125
dnn_sa_reverse	1.0	2.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.573 ± 0.268
llvm	1.0	2.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.600 ± 0.298
llvm_reverse	1.0	2.0	3.0	0.229 ± 0.084	0.405 ± 0.243	0.596 ± 0.316
lrzip	1.0	2.0	3.0	0.244 ± 0.118	0.367 ± 0.136	0.613 ± 0.228
lrzip_reverse	1.0	2.0	2.0	0.001 ± 0.002	0.001 ± 0.002	0.595 ± 0.314
mariadb	1.0	1.0	1.0	0.263 ± 0.423	0.264 ± 0.425	0.360 ± 0.466
mariadb_reverse	1.0	3.0	2.0	0.026 ± 0.083	0.318 ± 0.454	0.053 ± 0.111
mongodb	2.0	1.0	3.0	0.128 ± 0.166	0.000 ± 0.000	0.292 ± 0.350
mongodb_reverse	2.0	1.0	3.0	0.128 ± 0.135	0.026 ± 0.081	0.314 ± 0.404
storm_rs	1.0	2.0	2.0	0.058 ± 0.184	0.000 ± 0.000	0.525 ± 0.458
storm_rs_reverse	1.0	3.0	2.0	0.015 ± 0.047	0.247 ± 0.337	0.227 ± 0.173
storm_wc	1.0	2.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.406 ± 0.295
storm_wc_reverse	1.0	2.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.738 ± 0.240
trimesh	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
trimesh_reverse	1.0	3.0	2.0	0.026 ± 0.014	0.613 ± 0.226	0.337 ± 0.113
vp9	1.0	2.0	3.0	0.014 ± 0.046	0.043 ± 0.070	0.346 ± 0.284
vp9_reverse	2.0	2.0	1.0	0.098 ± 0.131	0.207 ± 0.298	0.152 ± 0.209
x264	1.0	2.0	1.0	0.500 ± 0.000	$\times$	0.500 ± 0.000
x264_reverse	1.0	2.0	1.0	0.500 ± 0.000	$\times$	0.500 ± 0.000
Average	1.19	1.96	2.04	0.148 ± 0.101	0.280 ± 0.173	0.424 ± 0.229

TABLE IV: Detailed case-level results of MMO_b , GA, and RS for software effort estimation. The formate is the same as Table I.

case	Rank			Mean $\pm$ Std		
	MMO_b	GA	RS	MMO_b	GA	RS
china_fold1	1.0	2.0	3.0	0.097 ± 0.074	0.160 ± 0.135	0.709 ± 0.168
china_fold2	1.0	2.0	3.0	0.027 ± 0.024	0.053 ± 0.055	0.803 ± 0.190
china_fold3	2.0	1.0	2.0	0.016 ± 0.024	0.016 ± 0.028	0.586 ± 0.248
desharnais_fold1	1.0	2.0	3.0	0.269 ± 0.188	0.396 ± 0.244	0.772 ± 0.154
desharnais_fold2	1.0	2.0	1.0	0.092 ± 0.067	0.242 ± 0.327	0.257 ± 0.076
desharnais_fold3	1.0	2.0	3.0	0.128 ± 0.104	0.332 ± 0.090	0.788 ± 0.149
finnish_fold1	2.0	1.0	3.0	0.127 ± 0.049	0.082 ± 0.072	0.541 ± 0.231
finnish_fold2	1.0	2.0	3.0	0.136 ± 0.095	0.172 ± 0.113	0.782 ± 0.141
finnish_fold3	2.0	1.0	2.0	0.025 ± 0.026	0.028 ± 0.026	0.711 ± 0.159
maxwell_fold1	2.0	1.0	3.0	0.079 ± 0.027	0.035 ± 0.028	0.907 ± 0.055
maxwell_fold2	2.0	1.0	3.0	0.074 ± 0.050	0.060 ± 0.045	0.943 ± 0.034
maxwell_fold3	2.0	1.0	3.0	0.136 ± 0.088	0.057 ± 0.034	0.899 ± 0.070
miyazaki_fold1	2.0	1.0	2.0	0.006 ± 0.004	0.006 ± 0.005	0.500 ± 0.242
miyazaki_fold2	2.0	1.0	2.0	0.005 ± 0.004	0.005 ± 0.003	0.481 ± 0.252
miyazaki_fold3	2.0	1.0	3.0	0.004 ± 0.002	0.003 ± 0.003	0.435 ± 0.261
Average	1.60	1.40	2.60	0.081 ± 0.055	0.110 ± 0.081	0.674 ± 0.162

TABLE V: Detailed case-level results of MMO_b , SBWOA, and RS for software defect prediction. The formate is the same as Table I.

case	Rank			Mean $\pm$ Std		
	MMO_b	SBWOA	RS	MMO_b	SBWOA	RS
ant-1.7_J48	1.0	1.0	2.0	0.122 ± 0.081	0.555 ± 0.196	0.604 ± 0.222
ant-1.7_KNN	1.0	1.0	2.0	0.128 ± 0.062	0.444 ± 0.124	0.466 ± 0.233
ant-1.7_LR	1.0	1.0	2.0	0.437 ± 0.147	0.626 ± 0.288	0.648 ± 0.331
ant-1.7_NB	1.0	1.0	2.0	0.188 ± 0.103	0.457 ± 0.279	0.442 ± 0.246
camel-1.6_J48	1.0	2.0	3.0	0.102 ± 0.077	0.496 ± 0.175	0.560 ± 0.267
camel-1.6_KNN	1.0	3.0	2.0	0.152 ± 0.074	0.636 ± 0.272	0.537 ± 0.249
camel-1.6_LR	1.0	1.0	2.0	0.178 ± 0.101	0.445 ± 0.277	0.458 ± 0.334
camel-1.6_NB	1.0	1.0	2.0	0.393 ± 0.111	0.574 ± 0.233	0.509 ± 0.246
ivy-2.0_J48	1.0	1.0	2.0	0.299 ± 0.152	0.762 ± 0.141	0.787 ± 0.134
ivy-2.0_KNN	1.0	1.0	2.0	0.133 ± 0.084	0.537 ± 0.199	0.504 ± 0.239
ivy-2.0_LR	1.0	1.0	2.0	0.150 ± 0.066	0.453 ± 0.246	0.477 ± 0.259
ivy-2.0_NB	1.0	1.0	2.0	0.139 ± 0.060	0.501 ± 0.241	0.442 ± 0.165
jedit-4.0_J48	1.0	2.0	3.0	0.135 ± 0.081	0.644 ± 0.206	0.766 ± 0.208
jedit-4.0_KNN	1.0	1.0	2.0	0.192 ± 0.082	0.662 ± 0.183	0.628 ± 0.174
jedit-4.0_LR	1.0	1.0	2.0	0.140 ± 0.090	0.574 ± 0.212	0.592 ± 0.194
jedit-4.0_NB	1.0	1.0	2.0	0.136 ± 0.071	0.604 ± 0.228	0.550 ± 0.185
lucene-2.4_J48	1.0	1.0	2.0	0.156 ± 0.083	0.667 ± 0.190	0.647 ± 0.126
lucene-2.4_KNN	1.0	2.0	3.0	0.250 ± 0.128	0.621 ± 0.207	0.719 ± 0.228
lucene-2.4_LR	1.0	2.0	3.0	0.440 ± 0.087	0.567 ± 0.273	0.636 ± 0.267
lucene-2.4_NB	1.0	1.0	2.0	0.352 ± 0.074	0.508 ± 0.292	0.476 ± 0.256
poi-3.0_J48	1.0	3.0	2.0	0.150 ± 0.102	0.778 ± 0.185	0.583 ± 0.282
poi-3.0_KNN	1.0	2.0	3.0	0.239 ± 0.100	0.572 ± 0.269	0.623 ± 0.221
poi-3.0_LR	1.0	1.0	2.0	0.243 ± 0.103	0.466 ± 0.230	0.476 ± 0.150
poi-3.0_NB	1.0	3.0	2.0	0.261 ± 0.115	0.565 ± 0.215	0.508 ± 0.186
synapse-1.2_J48	1.0	2.0	3.0	0.111 ± 0.079	0.555 ± 0.204	0.661 ± 0.202
synapse-1.2_KNN	1.0	2.0	3.0	0.127 ± 0.098	0.574 ± 0.246	0.598 ± 0.206
synapse-1.2_LR	1.0	1.0	2.0	0.217 ± 0.119	0.584 ± 0.199	0.597 ± 0.212
synapse-1.2_NB	1.0	1.0	2.0	0.197 ± 0.117	0.553 ± 0.242	0.567 ± 0.217
velocity-1.6_J48	1.0	1.0	2.0	0.088 ± 0.056	0.516 ± 0.262	0.509 ± 0.127
velocity-1.6_KNN	1.0	2.0	3.0	0.123 ± 0.091	0.422 ± 0.246	0.462 ± 0.228
velocity-1.6_LR	1.0	1.0	2.0	0.112 ± 0.089	0.357 ± 0.279	0.358 ± 0.275
velocity-1.6_NB	1.0	2.0	1.0	0.423 ± 0.083	0.597 ± 0.280	0.437 ± 0.200
xalan-2.6_J48	1.0	3.0	2.0	0.110 ± 0.079	0.492 ± 0.207	0.420 ± 0.159
xalan-2.6_KNN	1.0	3.0	2.0	0.127 ± 0.117	0.620 ± 0.114	0.507 ± 0.284
xalan-2.6_LR	1.0	1.0	1.0	0.361 ± 0.130	0.426 ± 0.308	0.431 ± 0.297
xalan-2.6_NB	2.0	1.0	2.0	0.293 ± 0.121	0.370 ± 0.278	0.483 ± 0.269
xerces-1.4_J48	1.0	2.0	1.0	0.229 ± 0.100	0.534 ± 0.347	0.316 ± 0.276
xerces-1.4_KNN	1.0	1.0	2.0	0.191 ± 0.181	0.604 ± 0.232	0.568 ± 0.246
xerces-1.4_LR	1.0	1.0	2.0	0.456 ± 0.138	0.595 ± 0.309	0.563 ± 0.319
xerces-1.4_NB	1.0	3.0	2.0	0.272 ± 0.152	0.578 ± 0.338	0.464 ± 0.338
Average	1.02	1.55	2.12	0.214 ± 0.100	0.552 ± 0.236	0.539 ± 0.231

TABLE VI: Detailed case-level results of MMO_b , LS-Sampling-Plus (LSP), RS for software product lines testing. The formate is the same as Table I.

case	Rank			Mean $\pm$ Std		
	MMO_b	LSP	RS	MMO_b	LSP	RS
7z	1.0	2.0	3.0	0.010 ± 0.005	0.463 ± 0.176	0.640 ± 0.218
amazon	1.0	2.0	3.0	0.040 ± 0.021	0.229 ± 0.131	0.720 ± 0.130
berkeleydbc	1.0	3.0	2.0	0.030 ± 0.026	0.865 ± 0.162	0.610 ± 0.151
cocheecologico	1.0	2.0	3.0	0.054 ± 0.033	0.485 ± 0.205	0.825 ± 0.127
counterstrikesimplefeaturemodel	1.0	2.0	3.0	0.082 ± 0.079	0.686 ± 0.136	0.846 ± 0.115
drupal	1.0	2.0	3.0	0.064 ± 0.058	0.345 ± 0.075	0.871 ± 0.076
dssample	1.0	3.0	2.0	0.029 ± 0.028	0.560 ± 0.357	0.400 ± 0.057
dune	1.0	3.0	2.0	0.092 ± 0.052	0.915 ± 0.057	0.723 ± 0.051
electronicdrum	1.0	2.0	3.0	0.126 ± 0.059	0.778 ± 0.066	0.842 ± 0.103
hipacc	1.0	2.0	3.0	0.078 ± 0.045	0.433 ± 0.074	0.814 ± 0.127
javagc	1.0	2.0	3.0	0.032 ± 0.031	0.431 ± 0.063	0.887 ± 0.132
jhipster	1.0	2.0	3.0	0.086 ± 0.055	0.520 ± 0.186	0.906 ± 0.093
lrzip	1.0	2.0	1.0	0.000 ± 0.000	0.544 ± 0.385	0.006 ± 0.020
modeltransformation	1.0	2.0	3.0	0.111 ± 0.076	0.336 ± 0.073	0.878 ± 0.145
polly	1.0	2.0	3.0	0.102 ± 0.050	0.748 ± 0.134	0.792 ± 0.152
smarthomev2.2	1.0	2.0	3.0	0.078 ± 0.077	0.356 ± 0.042	0.865 ± 0.102
splssimuelesnpn	1.0	3.0	2.0	0.074 ± 0.043	0.824 ± 0.085	0.779 ± 0.120
videoplayer	1.0	2.0	3.0	0.096 ± 0.059	0.446 ± 0.072	0.919 ± 0.072
vp9	1.0	2.0	3.0	0.099 ± 0.043	0.562 ± 0.077	0.768 ± 0.096
webportal	1.0	2.0	3.0	0.054 ± 0.044	0.336 ± 0.035	0.896 ± 0.064
x264	1.0	3.0	2.0	0.053 ± 0.043	0.900 ± 0.062	0.629 ± 0.062
Average	1.00	2.24	2.67	0.066 ± 0.044	0.560 ± 0.126	0.744 ± 0.105

TABLE VII: Detailed case-level results of MMO_b , GP-ILS, RS for software project scheduling. The formate is the same as Table I.

case	Rank			Mean $\pm$ Std		
	MMO_b	GP-ILS	RS	MMO_b	GP-ILS	RS
10-10-skill-4-5	1.0	2.0	3.0	0.004 ± 0.002	0.093 ± 0.022	0.966 ± 0.027
10-10-skill-6-7	1.0	2.0	3.0	0.002 ± 0.002	0.058 ± 0.011	0.972 ± 0.024
10-15-skill-4-5	1.0	2.0	3.0	0.004 ± 0.004	0.102 ± 0.022	0.978 ± 0.018
10-15-skill-6-7	1.0	2.0	3.0	0.001 ± 0.001	0.080 ± 0.013	0.992 ± 0.005
10-5-skill-4-5	1.0	2.0	3.0	0.000 ± 0.000	0.012 ± 0.007	0.975 ± 0.020
10-5-skill-6-7	1.0	2.0	3.0	0.003 ± 0.003	0.062 ± 0.029	0.965 ± 0.028
20-10-skill-4-5	1.0	2.0	3.0	0.004 ± 0.003	0.088 ± 0.033	0.981 ± 0.014
20-10-skill-6-7	1.0	2.0	3.0	0.002 ± 0.001	0.069 ± 0.019	0.987 ± 0.007
20-15-skill-4-5	1.0	2.0	3.0	0.004 ± 0.002	0.100 ± 0.013	0.984 ± 0.009
20-15-skill-6-7	1.0	2.0	3.0	0.010 ± 0.007	0.149 ± 0.032	0.976 ± 0.023
20-5-skill-4-5	1.0	2.0	3.0	0.007 ± 0.005	0.142 ± 0.051	0.961 ± 0.025
20-5-skill-6-7	1.0	2.0	3.0	0.004 ± 0.003	0.080 ± 0.024	0.956 ± 0.033
30-10-skill-4-5	1.0	2.0	3.0	0.003 ± 0.002	0.074 ± 0.033	0.974 ± 0.014
30-10-skill-6-7	1.0	2.0	3.0	0.004 ± 0.003	0.133 ± 0.060	0.984 ± 0.010
30-15-skill-4-5	1.0	2.0	3.0	0.003 ± 0.005	0.101 ± 0.010	0.975 ± 0.025
30-15-skill-6-7	1.0	2.0	3.0	0.006 ± 0.006	0.121 ± 0.025	0.976 ± 0.020
30-5-skill-4-5	1.0	2.0	3.0	0.008 ± 0.005	0.252 ± 0.087	0.953 ± 0.042
30-5-skill-6-7	1.0	2.0	3.0	0.004 ± 0.008	0.276 ± 0.088	0.964 ± 0.023
Average	1.00	2.00	3.00	0.004 ± 0.003	0.111 ± 0.032	0.973 ± 0.020

TABLE VIII: Detailed case-level results of MMO_b , GA, and RS for third party library migration. The format is the same as Table I.

case	Rank			Mean $\pm$ Std		
	MMO_b	GA	RS	MMO_b	GA	RS
migrationRule10_ALL	2.0	2.0	1.0	0.100 ± 0.316	0.253 ± 0.356	0.000 ± 0.000
migrationRule10_CO	1.0	1.0	1.0	0.100 ± 0.316	0.000 ± 0.000	0.000 ± 0.000
migrationRule10_DS	1.0	2.0	1.0	0.100 ± 0.316	0.600 ± 0.516	0.000 ± 0.000
migrationRule10_MS	2.0	2.0	1.0	0.000 ± 0.000	0.200 ± 0.422	0.000 ± 0.000
migrationRule18_ALL	1.0	2.0	3.0	0.016 ± 0.052	0.180 ± 0.306	0.191 ± 0.140
migrationRule18_CO	1.0	1.0	1.0	0.100 ± 0.316	0.000 ± 0.000	0.000 ± 0.000
migrationRule18_DS	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
migrationRule18_MS	1.0	1.0	1.0	0.017 ± 0.053	0.117 ± 0.315	0.033 ± 0.070
migrationRule1_ALL	1.0	2.0	3.0	0.111 ± 0.068	0.221 ± 0.161	0.830 ± 0.116
migrationRule1_CO	1.0	2.0	3.0	0.086 ± 0.136	0.274 ± 0.224	0.926 ± 0.076
migrationRule1_DS	2.0	2.0	1.0	0.000 ± 0.000	0.200 ± 0.422	0.000 ± 0.000
migrationRule1_MS	2.0	1.0	2.0	0.029 ± 0.090	0.029 ± 0.090	0.871 ± 0.045
migrationRule2_ALL	1.0	2.0	3.0	0.074 ± 0.049	0.284 ± 0.119	0.934 ± 0.052
migrationRule2_CO	1.0	2.0	3.0	0.153 ± 0.129	0.260 ± 0.088	0.952 ± 0.031
migrationRule2_DS	1.0	2.0	3.0	0.074 ± 0.061	0.203 ± 0.119	0.938 ± 0.042
migrationRule2_MS	1.0	2.0	3.0	0.119 ± 0.064	0.271 ± 0.127	0.933 ± 0.046
migrationRule3_ALL	1.0	1.0	2.0	0.421 ± 0.454	0.482 ± 0.177	0.507 ± 0.211
migrationRule3_CO	2.0	2.0	1.0	0.485 ± 0.337	0.503 ± 0.264	0.383 ± 0.110
migrationRule3_MS	2.0	1.0	2.0	0.250 ± 0.354	0.250 ± 0.354	0.500 ± 0.000
migrationRule4_ALL	1.0	2.0	3.0	0.085 ± 0.082	0.166 ± 0.136	0.863 ± 0.117
migrationRule4_CO	1.0	2.0	3.0	0.000 ± 0.000	0.100 ± 0.211	0.800 ± 0.258
migrationRule4_DS	2.0	1.0	2.0	0.045 ± 0.144	0.055 ± 0.172	0.815 ± 0.188
migrationRule4_MS	2.0	1.0	2.0	0.000 ± 0.000	0.000 ± 0.000	0.833 ± 0.112
migrationRule5_ALL	1.0	1.0	2.0	0.290 ± 0.362	0.420 ± 0.237	0.466 ± 0.122
migrationRule5_CO	2.0	1.0	2.0	0.233 ± 0.225	0.333 ± 0.385	0.600 ± 0.263
migrationRule5_DS	1.0	2.0	3.0	0.190 ± 0.402	0.551 ± 0.475	0.903 ± 0.000
migrationRule5_MS	1.0	1.0	1.0	0.100 ± 0.316	0.100 ± 0.316	0.200 ± 0.422
migrationRule7_ALL	1.0	2.0	3.0	0.137 ± 0.102	0.229 ± 0.152	0.886 ± 0.082
migrationRule7_CO	1.0	2.0	3.0	0.183 ± 0.130	0.245 ± 0.157	0.879 ± 0.081
migrationRule7_DS	1.0	2.0	3.0	0.152 ± 0.077	0.231 ± 0.098	0.884 ± 0.070
migrationRule7_MS	1.0	2.0	3.0	0.225 ± 0.162	0.481 ± 0.241	0.894 ± 0.084
migrationRule8_ALL	1.0	2.0	3.0	0.000 ± 0.000	0.100 ± 0.211	0.729 ± 0.312
migrationRule8_CO	2.0	1.0	2.0	0.000 ± 0.000	0.044 ± 0.141	0.478 ± 0.240
migrationRule8_DS	1.0	1.0	1.0	0.500 ± 0.000	0.500 ± 0.000	0.500 ± 0.000
migrationRule8_MS	2.0	1.0	2.0	0.000 ± 0.000	0.020 ± 0.063	0.560 ± 0.207
Average	1.31	1.57	2.11	0.139 ± 0.146	0.240 ± 0.202	0.565 ± 0.100

TABLE IX: Detailed case-level results of MMO_b , HGA, and RS for workflow scheduling. The format is the same as Table I.

case	Rank			Mean $\pm$ Std		
	MMO_b	HGA	RS	MMO_b	HGA	RS
CyberShake_100	1.0	2.0	3.0	0.186 ± 0.066	0.377 ± 0.280	0.569 ± 0.107
CyberShake_1000	2.0	1.0	3.0	0.336 ± 0.139	0.260 ± 0.161	0.925 ± 0.056
CyberShake_30	1.0	3.0	2.0	0.068 ± 0.066	0.317 ± 0.299	0.153 ± 0.190
CyberShake_50	1.0	3.0	2.0	0.087 ± 0.062	0.342 ± 0.257	0.230 ± 0.046
Epigenomics_100	1.0	2.0	3.0	0.376 ± 0.274	0.527 ± 0.275	0.765 ± 0.102
Epigenomics_24	1.0	3.0	2.0	0.197 ± 0.090	0.574 ± 0.225	0.357 ± 0.091
Epigenomics_46	1.0	3.0	2.0	0.141 ± 0.150	0.630 ± 0.271	0.604 ± 0.128
Epigenomics_997	2.0	1.0	3.0	0.524 ± 0.214	0.211 ± 0.171	0.733 ± 0.106
Inspirai_100	1.0	2.0	3.0	0.163 ± 0.147	0.466 ± 0.170	0.834 ± 0.124
Inspirai_1000	2.0	1.0	3.0	0.434 ± 0.157	0.245 ± 0.189	0.866 ± 0.089
Inspirai_30	1.0	2.0	3.0	0.001 ± 0.000	0.165 ± 0.313	0.282 ± 0.268
Inspirai_50	1.0	2.0	3.0	0.118 ± 0.136	0.636 ± 0.177	0.839 ± 0.122
Montage_100	1.0	2.0	3.0	0.107 ± 0.076	0.244 ± 0.089	0.721 ± 0.201
Montage_1000	2.0	1.0	3.0	0.248 ± 0.093	0.178 ± 0.154	0.883 ± 0.092
Montage_25	1.0	3.0	2.0	0.064 ± 0.145	0.641 ± 0.156	0.547 ± 0.024
Montage_50	1.0	2.0	3.0	0.088 ± 0.068	0.415 ± 0.239	0.510 ± 0.106
Sipht_100	1.0	3.0	2.0	0.073 ± 0.085	0.304 ± 0.360	0.115 ± 0.065
Sipht_1000	2.0	1.0	2.0	0.164 ± 0.113	0.174 ± 0.162	0.568 ± 0.322
Sipht_30	1.0	3.0	2.0	0.081 ± 0.057	0.754 ± 0.250	0.609 ± 0.136
Sipht_60	1.0	3.0	2.0	0.036 ± 0.024	0.608 ± 0.456	0.092 ± 0.018
Average	1.25	2.15	2.55	0.175 ± 0.108	0.403 ± 0.233	0.560 ± 0.120

TABLE X: Detailed case-level results of MMO_b , HWOA, and RS for web service composition. The format is the same as Table I.

case	Rank			Mean $\pm$ Std		
	MMO_b	HWOA	RS	MMO_b	HWOA	RS
workflow_1	1.0	2.0	3.0	0.228 ± 0.170	0.443 ± 0.290	0.783 ± 0.118
workflow_10	2.0	1.0	3.0	0.481 ± 0.263	0.312 ± 0.203	0.587 ± 0.144
workflow_10_reverse	3.0	1.0	2.0	0.809 ± 0.149	0.513 ± 0.211	0.664 ± 0.154
workflow_1_reverse	3.0	1.0	2.0	0.711 ± 0.231	0.576 ± 0.339	0.682 ± 0.160
workflow_2	2.0	1.0	2.0	0.356 ± 0.213	0.293 ± 0.222	0.842 ± 0.161
workflow_2_reverse	2.0	1.0	3.0	0.781 ± 0.160	0.610 ± 0.293	0.843 ± 0.077
workflow_3	2.0	1.0	2.0	0.369 ± 0.299	0.390 ± 0.105	0.814 ± 0.111
workflow_3_reverse	2.0	1.0	3.0	0.747 ± 0.112	0.631 ± 0.296	0.778 ± 0.084
workflow_4	2.0	1.0	3.0	0.440 ± 0.272	0.228 ± 0.196	0.556 ± 0.169
workflow_4_reverse	1.0	3.0	2.0	0.596 ± 0.259	0.688 ± 0.252	0.676 ± 0.175
workflow_5	2.0	1.0	3.0	0.378 ± 0.198	0.240 ± 0.186	0.679 ± 0.255
workflow_5_reverse	3.0	1.0	2.0	0.607 ± 0.187	0.281 ± 0.215	0.531 ± 0.138
workflow_6	1.0	2.0	3.0	0.289 ± 0.291	0.317 ± 0.207	0.726 ± 0.155
workflow_6_reverse	1.0	1.0	2.0	0.562 ± 0.291	0.493 ± 0.232	0.572 ± 0.124
workflow_7	2.0	1.0	2.0	0.471 ± 0.330	0.400 ± 0.231	0.824 ± 0.124
workflow_7_reverse	3.0	1.0	2.0	0.574 ± 0.234	0.458 ± 0.246	0.547 ± 0.085
workflow_8	2.0	1.0	3.0	0.587 ± 0.280	0.165 ± 0.163	0.735 ± 0.162
workflow_8_reverse	1.0	1.0	2.0	0.598 ± 0.291	0.419 ± 0.193	0.561 ± 0.129
workflow_9	2.0	1.0	3.0	0.404 ± 0.256	0.232 ± 0.197	0.733 ± 0.208
workflow_9_reverse	1.0	1.0	2.0	0.622 ± 0.297	0.377 ± 0.265	0.672 ± 0.148
Average	1.90	1.20	2.45	0.530 ± 0.239	0.403 ± 0.227	0.690 ± 0.144